Home About Contact

Sales and Revenue

PIZZA HUB

- DATA ANALYSIS USING SQL

Home About Contact

PROJECT DESCRIPTION

PIZZAHUB SALES AND REVENUE ANALYSIS USING SQL

Performed end-to-end sales data analysis on PizzaHub's database using SQL. Solved business-focused queries ranging from basic sales metrics to advanced revenue analysis. Extracted insights on order trends, customer preferences, pizza category performance, revenue contribution, peak ordering hours, and cumulative sales growth using joins, aggregations, subqueries, CTEs, and window functions.

Home About Contact

PIZZAHUB SALES AND REVENUE ANALYSIS USING SQL

Analyzed PizzaHub sales data using SQL to uncover insights into sales performance, revenue generation, customer ordering patterns, and product popularity. Performed data analysis to identify top-selling pizzas, category-wise sales distribution, peak ordering hours, revenue contribution, and cumulative revenue trends. Utilized SQL operations including SELECT, FROM, JOIN, GROUP BY, ORDER BY, SUM(), COUNT(), ROUND(), OVER(), RANK(), LIMIT, and subqueries to solve business-oriented analytical problems.

Tools: MySQL, SQL

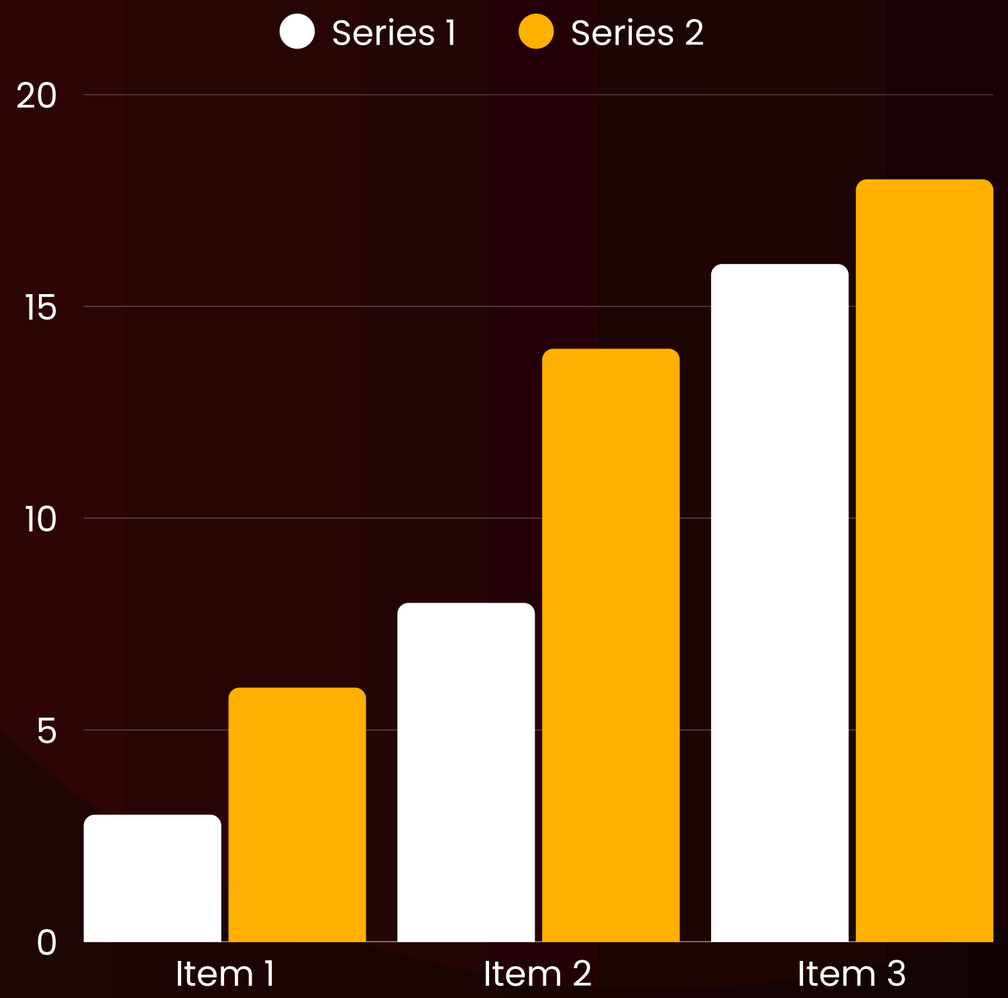
Skills: Data Analysis, Joins, Aggregate Functions, Window Functions, Ranking Functions, Business Intelligence Reporting.

Home About Contact

SALES REPORT

ANALYSIS

This report presents an analysis of PizzaHub sales data using SQL. The project focuses on evaluating sales performance, revenue generation, customer ordering behavior, and product popularity. Through various SQL queries, key business insights were extracted, including top-selling pizzas, category-wise sales distribution, peak ordering hours, and revenue trends. The findings demonstrate how data-driven analysis can support better business decisions and improve operational performance.



Home About Contact

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```
SELECT
    COUNT(order_id)
FROM
    orders;
```

Result Grid

	COUNT(order_id)
▶	21350

Home About Contact

CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT
    ROUND(SUM(orders_details.quantity * pizzas.price),
            2) AS total_sales
FROM
    orders_details
        JOIN
    pizzas ON pizzas.pizza_id = orders_details.pizza_id;
```

Result Grid

total_sales
817860.05

IDENTIFY THE HIGHEST-PRICED PIZZA.

```sql
SELECT
    pizza_types.name, pizzas.price AS highest_priced
FROM
    pizzas
        JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid | Filter Rows:

name	highest_priced
The Greek Pizza	35.95

Home About Contact

IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
    pizzas.size, COUNT(orders_details.order_details_id) AS order_size
FROM
    pizzas
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_size DESC;
```

Result Grid

size	order_size
L	18526
M	15385
S	14137
XL	544
XXL	28

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITY

```sql
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS quantity
FROM
    pizzas
        JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows:

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```sql
SELECT
    pizza_types.category,
    SUM(orders_details.quantity) AS total_category
FROM
    pizzas
        JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizza_types.category
ORDER BY total_category desc;
```

Result Grid | Filter Rows

category	total_category
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Home About Contact

DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```sql
SELECT
    HOUR(order_time) AS hours, COUNT(order_id) AS counts
FROM
    orders
GROUP BY hours;
```

Result Grid

hours	counts
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS

```sql
SELECT
    category, COUNT(name) AS cat_distr
FROM
    pizza_types
GROUP BY category;
```

Result Grid

category	cat_distr
Chicken	6
Classic	8
Supreme	9
Veggie	9

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```sql
SELECT
    ROUND(AVG(no_of_order), 0) AS avg_piz_per_day
FROM
    (SELECT
        orders.order_date AS order_day,
            SUM(orders_details.quantity) AS no_of_order
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date) AS order_num;
```

Home About Contact

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```sql
SELECT
    pizza_types.name,
    SUM(pizzas.price * orders_details.quantity) AS rev
FROM
    orders_details
        JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
        JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.name
ORDER BY rev DESC
LIMIT 3;
```

Result Grid | Filter Rows:

name	rev
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Home About Contact

CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```sql
SELECT
    pizza_types.category,
    ROUND((SUM(pizzas.price * orders_details.quantity) / (SELECT
                    ROUND(SUM(orders_details.quantity * pizzas.price),
                                2) AS total_sales
                FROM
                    orders_details
                        JOIN
                    pizzas ON pizzas.pizza_id = orders_details.pizza_id)) * 100,
            2) AS tot_rev
FROM
    orders_details
        JOIN
    pizzas ON orders_details.pizza_id = pizzas.pizza_id
        JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.category
ORDER BY tot_rev DESC;
```

Result Grid

category	tot_rev
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

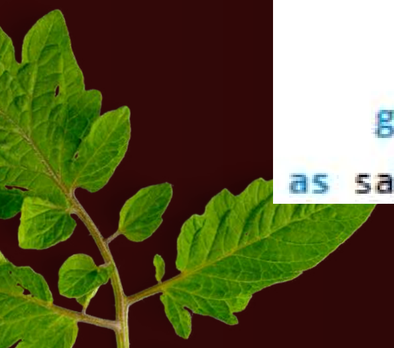
Home About Contact

ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
select order_date,
    sum(revn) over(order by order_date) as rev_cumm
from
    (select orders.order_date,
        sum(orders_details.quantity* pizzas.price)
            as revn
    from orders join orders_details on
        orders.order_id  = orders_details.order_id
            join pizzas on
        pizzas.pizza_id = orders_details.pizza_id
    group by orders.order_date)
as sales;
```

Result Grid | Filter Rows:

order_date	rev_cumm
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004
2015-01-15	34343.50000000001
2015-01-16	36937.65000000001
2015-01-17	39001.75000000001
2015-01-18	40978.600000000006
2015-01-19	43365.75000000001
2015-01-20	45763.65000000001
2015-01-21	47804.20000000001
2015-01-22	50300.90000000001
2015-01-23	52724.600000000006
2015-01-24	55013.850000000006
2015-01-25	56631.40000000001



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```sql
select name, revenue from
    (select category,name , revenue,
    rank() over(partition by category order by revenue desc) as rn
from
    (select pizza_types.category, pizza_types.name,
    sum(orders_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas
    on pizza_types.pizza_type_id = pizzas.pizza_type_id
    join orders_details
    on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name ) as a)as b
where rn <= 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000065
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

CONCLUSION

- Successfully analyzed PizzaHub sales data using SQL.
- Identified key trends in customer orders, revenue generation, and product performance.
- Determined top-selling pizzas, popular categories, and peak ordering periods.
- Applied SQL techniques such as joins, aggregations, subqueries, and window functions to extract meaningful business insights.
- Demonstrated how data analysis can support informed business decisions and improve overall sales performance.

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THANK YOU

SHUBHANGI SONKAR

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